

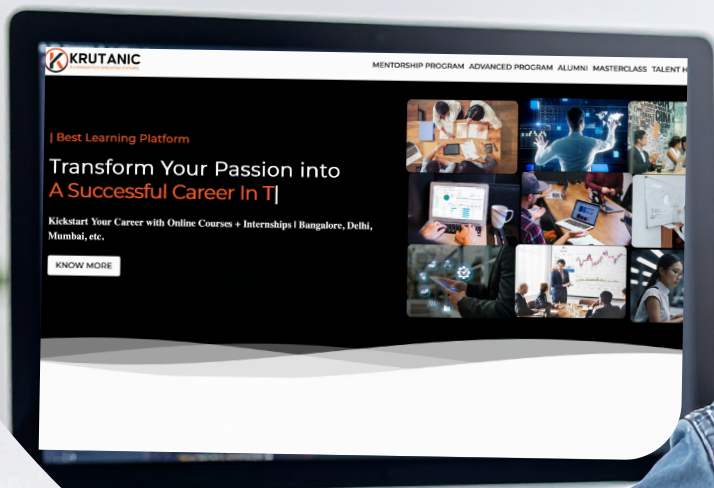


KRUTANIC

A LADDER FOR BRIGHTER FUTURE

*Transform Your Passion into
A Successful Career In Tech*

EMBEDDED SYSTEMS



We are now an **accredited**
partner under



www.krutanic.com



support@krutanic.com



EMBEDDED SYSTEMS

ABOUT US

Krutanic is at the forefront of transforming education through cutting-edge technology. Our comprehensive platform empowers learners with personalized learning experiences, collaborative tools, and real-time analytics. With adaptive assessments and interactive content creation, we enhance student engagement and achievement. Join us in revolutionizing education for the digital age, driving positive outcomes and preparing learners for success in tomorrow's world.

TRAINING AND INTERNSHIP PROGRAM

Why Embedded Systems?

- Power core technologies like IoT, robotics, and automation
- Essential in consumer electronics, automotive, aerospace, and healthcare
- High demand for skilled professionals across industries
- Offers a blend of hardware and software skills
- Enables real-time, reliable, and efficient system performance
- Critical for innovation in smart devices and connected systems

TRAINING AND INTERNSHIP PROGRAM

1st MONTH



LIVE SESSIONS

Live sessions with industrial experts having experience above 10 years in the industry.



RECORDINGS

Recordings of all live sessions available with 1 year access in our LMS portal.



INDUSTRY RELATED

Industry related curriculum designed by the working professionals in the top hierarchy.



TRAINING AND INTERNSHIP PROGRAM

nd
2 MONTH

REAL TIME PROJECTS

Two real time industrial projects
One minor project and One major project.



ASSISTANCE

All mentors will be assigned as project leads and guide the intern till the completion of the project.

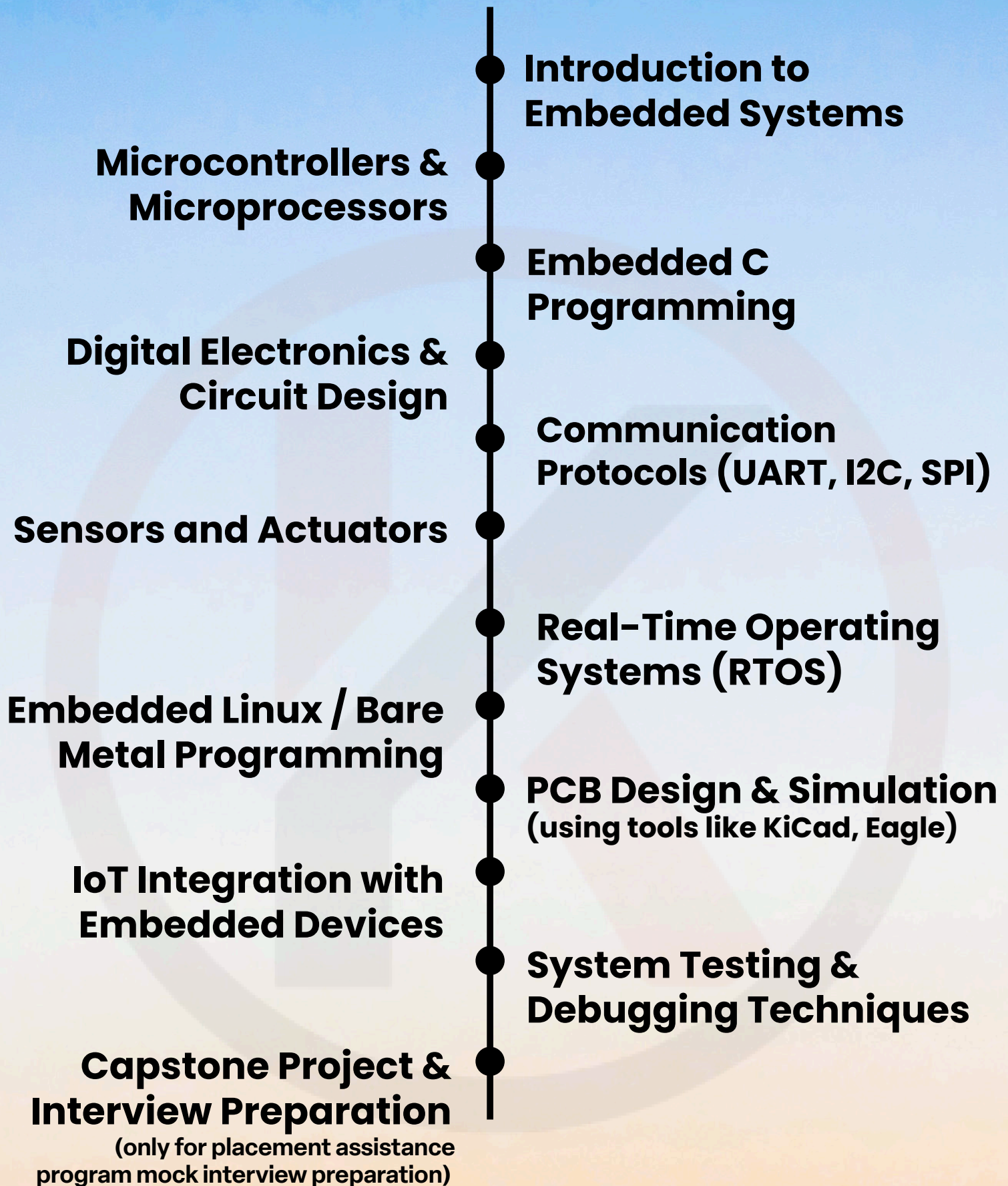


PERSONAL DEVELOPMENT

Additional projects for personal development can be required.



Modules Overview





Curriculum included

1

Introduction to Embedded Systems

- Understand the definition and real-world applications
- Explore system components: hardware, software, and firmware
- Learn about embedded constraints (power, memory, performance)

2

Microcontrollers & Microprocessors

- Understand the architecture and key differences
- Explore examples like 8051, ARM, AVR, and Arduino
- Learn how to select a microcontroller for a project

3

Embedded C Programming

- Learn data types, loops, conditionals, and functions in C
- Interface peripherals like LEDs, motors, and sensors
- Understand memory management and register-level programming

4

Digital Electronics & Circuit Design

- Basics of logic gates, multiplexers, flip-flops, counters
- Learn circuit design techniques and boolean algebra
- Work on breadboards and simulation tools

Curriculum included

5

Communication Protocols (UART, I2C, SPI, CAN)

- Study synchronous and asynchronous communication
- Interface devices using serial protocols
- Understand timing, addressing, and data transfer

6

Sensors and Actuators

- Learn about analog and digital sensors
- Interface actuators like servos, motors, and buzzers
- Read sensor data and convert it for microcontroller input

7

Real-Time Operating Systems (RTOS)

- Understand tasks, scheduling, semaphores, and queues
- Learn multitasking in embedded systems
- Implement RTOS on microcontrollers

8

Embedded Linux / Bare Metal Programming

- Explore Linux kernel, bootloaders, and drivers
- Understand file systems, GPIO access, and system calls
- Learn bare-metal coding without an OS for speed-critical applications

Curriculum included

9

PCB Design & Simulation (using tools like KiCad, Eagle)

- Schematic capture and layout design
- Place components and route tracks
- Run simulations and generate Gerber files

10

IoT Integration with Embedded Devices

- Interface Wi-Fi/Bluetooth modules (e.g., ESP8266, ESP32)
- Send data to cloud platforms like ThingSpeak or Firebase
- Control devices remotely via mobile apps or web servers

11

System Testing & Debugging Techniques

- Use tools like oscilloscopes, logic analyzers, and multimeters
- Debug embedded C code using breakpoints and watches
- Perform unit, integration, and system-level testing

12

Capstone Project & Interview Preparation

- Build a real-world embedded system project
- Document the design, code, and outcomes
- Prepare for technical interviews and coding rounds

Tools Included



Ardutno



XIDE



FREE RTOS



KEIL

Discover why this **Embedded Systems** course is essential for your learning journey.

STEP 1



Google
Ratings 4.8/5

STEP 2



Mentees
Trained 15k+

STEP 3



Hiring
Partners 200+

STEP 6



Global Size USD
200 BILLION

STEP 5



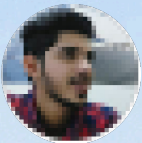
Average Salary
10+ LPA

STEP 4



Job Openings
25000+

Students Reviews



Ramshad K

★★★★★

I have gained a really hands on Experience with on graphic design which was delivered by Akash sir was very supportive and endless worthy experience so far.



Rajendra Prasad

★★★★★

I'm Rajendran Prasad, to be honest I'm expressing my words, I have researched and found the correct certificate on LinkedIn and scan that and found it 100% genuine and verified successfully.

It's my individual recommendation for everyone that Krutanic company is perfect with quality training and real world projects.

I have seen some bad reviews and many good reviews so got confused but now I realised that Krutanic solutions is the best company ever for training and internship provider, I have learned a lot of new industrial things.



Sahad K

★★★★★

I got placement at Wipro through krutanic support team. I am really grateful that I got to learn from AI Mentor Ashish sir and support from my counsellor was really helpful. All thanks to Bhoomika mam my operations support member from krutanic.



Sunil Kumar

★★★★★

I'm Sunil Kumar, I have enrolled for Data Science program at Krutanic solutions and my experience is outstanding.

It's my individual recommendation for everyone that Krutanic company is perfect with quality training and real world projects.



Shubham Rai

★★★★★

Krutanic Group is truly doing a remarkable job by creating learning spaces that are not only informative but also inspiring. I've had the opportunity to attend two of their courses-Photography and Graphic Designing-and both were simply



Aman Nema

★★★★★

Krutanics Full Stack Web Development program was a fantastic learning experience. The course covered essential technologies like HTML, CSS, JavaScript, React, and backend development, providing strong foundation for building real-world web applications. The hands-on projects and expert guidance from mentors made learning interactive and practical. This program is perfect for beginners and aspiring developers looking to enhance their skills in web development. Highly recommended.

Students Reviews



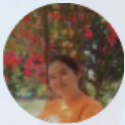
Tejas Kolekar

I am thrilled to share my enriching experience from the Artificial Intelligence and Machine Learning Internship I recently completed with Krutanic, in collaboration with Wipro, a leading multinational corporation. This 3-month program was an incredible opportunity to deepen my knowledge, enhance my skills, and grow both personally and professionally in the dynamic field of AI and ML.



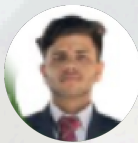
Pranali Shinde

Excited to share that I've just completed a Data Science course from Krutanic! It was an amazing journey diving into data visualization, exploratory data analysis, SQL, and machine learning. I've gained some valuable skills that I can't wait to apply in real-world projects. Big thanks to the instructors and everyone who made this experience so enriching! Looking forward to what's next!



Prabhleen Kaur

Just after few months of completion of my AI course I got my hands on attending Master class on Building AI agent without coding held in the month of March my experience was very good I get to know about new things from my mentor Mr.Aashish Mishra And at the end i recieved this ©
All thanks to Krutanic for giving me this opportunity.....



Rishabh Nagar

Certificate of Training Completion from Krutanic Solutions in the field of Android App Development Krutanic Paidi Venkata
[#krutanic](#) [#android](#) [#app](#)
[#development](#) [#androidstudio](#)



Himendra Shukla

Completing my internship at **KrutaniC** has been a transformative journey, equipping me with hands-on experience in **Full Stack Web Development** and preparing me for a bright future in tech. Grateful for the guidance, real-world projects, and skills gained — here's to innovation and success ahead! ★
[#InternshipCompletion](#) [#FullStackDevelopment](#) [#KrutaniC](#)
[#TechJourney](#)
Krutanic



Touheed Ahmed Nadaf

Clouds conquered.
Excited to share my Cloud Computing internship certification from Krutanic!
Grateful for the hands-on learning and ready to turn this experience into an impact in the ever-evolving tech landscape.
[#CloudComputing](#) [#Krutanic](#) [#InternshipWin](#)
[#TechReady](#) [#NextGenTech](#)
[#FutureInCloud](#) [#LevelUp](#) [#DigitalSkills](#)

Career Opportunities in Embedded Systems

Embedded Systems Engineer Design and develop embedded hardware and software for various electronic devices.

Firmware Developer Write low-level code that directly controls hardware components.

Embedded Software Developer Develop applications and drivers that run on microcontrollers or embedded processors.

IoT Developer Build Internet of Things devices by integrating sensors, actuators, and embedded platforms.

Hardware Design Engineer Design PCBs (Printed Circuit Boards) and integrate microcontrollers and peripherals.

System Validation & Testing Engineer Test embedded systems for performance, reliability, and compliance.

Certifications

01

**Training
Completion
Certificate of
Validates the skills
which acquired**

02

**Internship
Completion
certificate
certified by
Adobe**

04

**Certificate of
excellence
(Based on you
Performance)**

03

**Letter of
recommendation(L
OR) for your
JOB/PLACEMENT**

STUDENT AVERAGE RATINGS

★★★★★ 4.85/5

CERTIFICATE OF TRAINING COMPLETION

THIS CERTIFICATE IS PROUDLY PRESENTED TO

Danish Raja Akhtar

Your dedication and achievement are truly inspiring, reflecting your readiness to excel in the dynamic field. This Certificate of Training Completion is awarded for successfully completing the Full Stack Web Development on May 2025



Shrikant
SHRIKANT RATHOD
EXECUTIVE DIRECTOR



CERTIFICATE OF INTERNSHIP

May 5th, 2025 to July 5th, 2025

This Certificate is Proudly Presented To

Danish Raja Akhtar

This is to certify that the candidate has successfully completed the Internship. During the internship period, they demonstrated strong technical skills, professionalism, and a keen interest in the field. Their contributions were valuable to the team in the domain of Full Stack Web Development



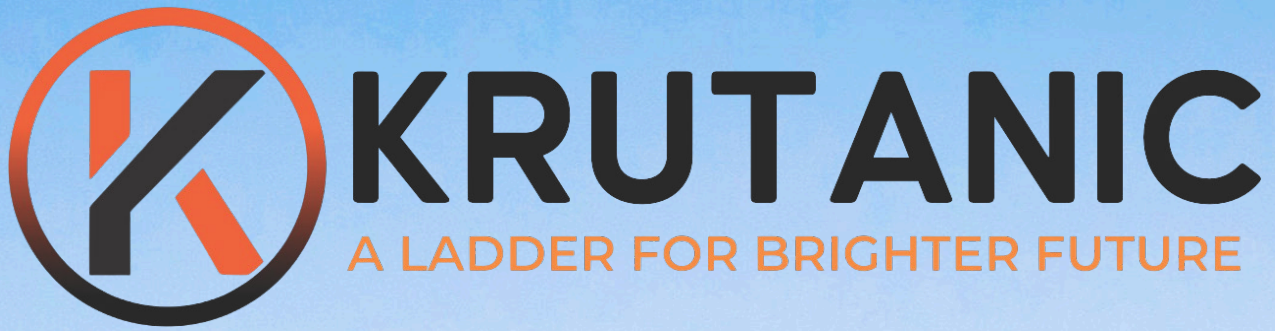
<https://www.krutanic.com/verify>
USER VERIFICATION CODE: 6880446cbf6aa824890024f3

ENROLLMENT VERIFICATION CODE: KRT_5678



Shrikant
Shrikant Rathod
Executive Director





Companies where our alumni thrive



Come join us now
and contact us now



+ 91 7022812878

www.krutanic.com

Bangalore, Karnataka

Our process

Quick Guide

Register With Us

Schedule Your
Class

Certifications